

```
In [1]: import os
os.getcwd()
```

Out[1]: 'C:\\Users\\VICTUS'

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

```
In [3]: df = pd.read_csv("customer_behavior_analysis.csv")

df.head()
```

Out[3]:

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns
0	44605	2023-05-03 21:30:02	Home	177	1	2427	PayPal	31	
1	44605	2021-05-16 13:57:44	Electronics	174	3	2448	PayPal	31	
2	44605	2020-07-13 06:16:57	Books	413	1	2345	Credit Card	31	
3	44605	2023-01-17 13:14:36	Electronics	396	3	937	Cash	31	
4	44605	2021-05-01 11:29:27	Books	259	4	2598	PayPal	31	

```
In [4]: df.columns
```

Out[4]: Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price', 'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age', 'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'], dtype='str')

```
In [5]: df.info()
```

```

<class 'pandas.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 13 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   Customer ID           250000 non-null  int64  
 1   Purchase Date         250000 non-null  str     
 2   Product Category      250000 non-null  str     
 3   Product Price         250000 non-null  int64  
 4   Quantity              250000 non-null  int64  
 5   Total Purchase Amount 250000 non-null  int64  
 6   Payment Method        250000 non-null  str     
 7   Customer Age          250000 non-null  int64  
 8   Returns               202618 non-null  float64 
 9   Customer Name         250000 non-null  str     
10   Age                   250000 non-null  int64  
11   Gender                250000 non-null  str     
12   Churn                 250000 non-null  int64  
dtypes: float64(1), int64(7), str(5)
memory usage: 37.0 MB

```

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In [6]: df.isnull().sum()
```

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Out[6]: Customer ID           0
        Purchase Date        0
        Product Category      0
        Product Price         0
        Quantity              0
        Total Purchase Amount  0
        Payment Method        0
        Customer Age          0
        Returns               47382
        Customer Name         0
        Age                   0
        Gender                0
        Churn                 0
        dtype: int64

```

```
In [7]: df.describe()
```

Out[7]:

	Customer ID	Product Price	Quantity	Total Purchase Amount	Customer Age	Returns
count	250000.000000	250000.000000	250000.000000	250000.000000	250000.000000	202618.
mean	25017.632092	254.742724	3.004936	2725.385196	43.798276	0.
std	14412.515718	141.738104	1.414737	1442.576095	15.364915	0.
min	1.000000	10.000000	1.000000	100.000000	18.000000	0.
25%	12590.000000	132.000000	2.000000	1476.000000	30.000000	0.
50%	25011.000000	255.000000	3.000000	2725.000000	44.000000	1.
75%	37441.250000	377.000000	4.000000	3975.000000	57.000000	1.
max	50000.000000	500.000000	5.000000	5350.000000	70.000000	1.

```
In [8]: df.median(numeric_only=True)
```

Out[8]: Customer ID 25011.0
Product Price 255.0
Quantity 3.0
Total Purchase Amount 2725.0
Customer Age 44.0
Returns 1.0
Age 44.0
Churn 0.0
dtype: float64

```
In [9]: df.mode().iloc[0]
```

Out[9]: Customer ID 48382.0
Purchase Date 2020-01-01 15:49:02
Product Category Electronics
Product Price 290.0
Quantity 4.0
Total Purchase Amount 2533.0
Payment Method Credit Card
Customer Age 47.0
Returns 1.0
Customer Name Michael Johnson
Age 47.0
Gender Male
Churn 0.0
Name: 0, dtype: object

```
In [10]: df = df.dropna()
```

```
In [11]: df = df.drop_duplicates()
```

```
In [14]: df.columns
```

```
Out[14]: Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',
               'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',
               'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],
              dtype='str')
```

```
In [15]: df['Total_Spending'] = df['Quantity'] * df['Product Price']
```

```
In [16]: df['Purchase_Date'] = pd.to_datetime(df['Purchase Date'])
```

```
In [ ]:
```

```
In [23]: import pandas as pd
import datetime as dt

# Convert Purchase Date column to datetime
df['Purchase Date'] = pd.to_datetime(df['Purchase Date'])

# Current date
today = dt.datetime.now()

# Create RFM table
rfm = df.groupby('Customer ID').agg({
    'Purchase Date': lambda x: (today - x.max()).days,
    'Payment Method': 'count',
    'Total_Spending': 'sum'
})

# Rename columns
rfm.columns = ['Recency', 'Frequency', 'Monetary']

# Display output
rfm.head()
```

```
Out[23]:
```

	Recency	Frequency	Monetary
Customer ID			

Customer ID			
1	1871	1	2280
2	1055	6	6459
3	1205	1	564
4	1424	5	4339
5	1520	3	1230

```
In [24]: scaler = StandardScaler()

rfm_scaled = scaler.fit_transform(rfm)
```

```
In [25]: kmeans = KMeans(n_clusters=4, random_state=42)

rfm['Cluster'] = kmeans.fit_predict(rfm_scaled)
```

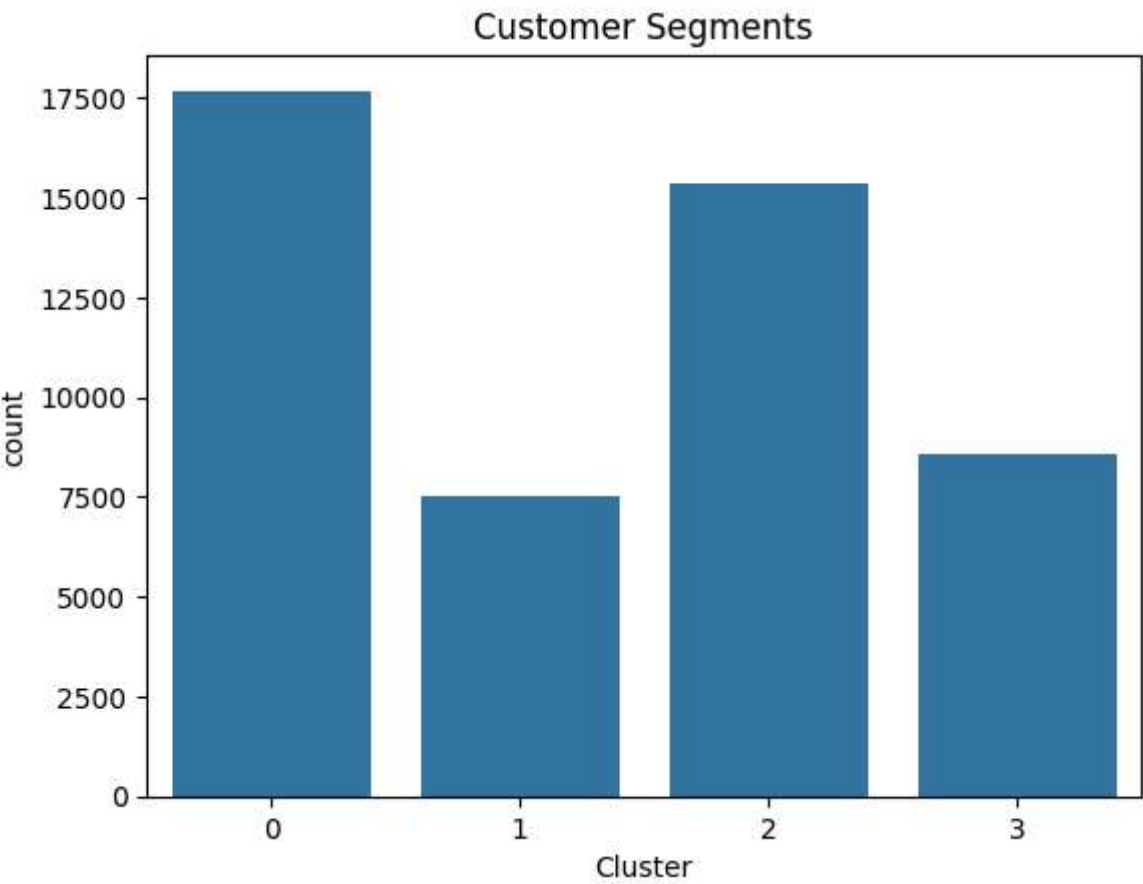
```
rfm.head()
```

Out[25]:

	Recency	Frequency	Monetary	Cluster
Customer ID				
1	1871	1	2280	3
2	1055	6	6459	1
3	1205	1	564	2
4	1424	5	4339	0
5	1520	3	1230	3

```
In [26]: sns.countplot(x='Cluster', data=rfm)

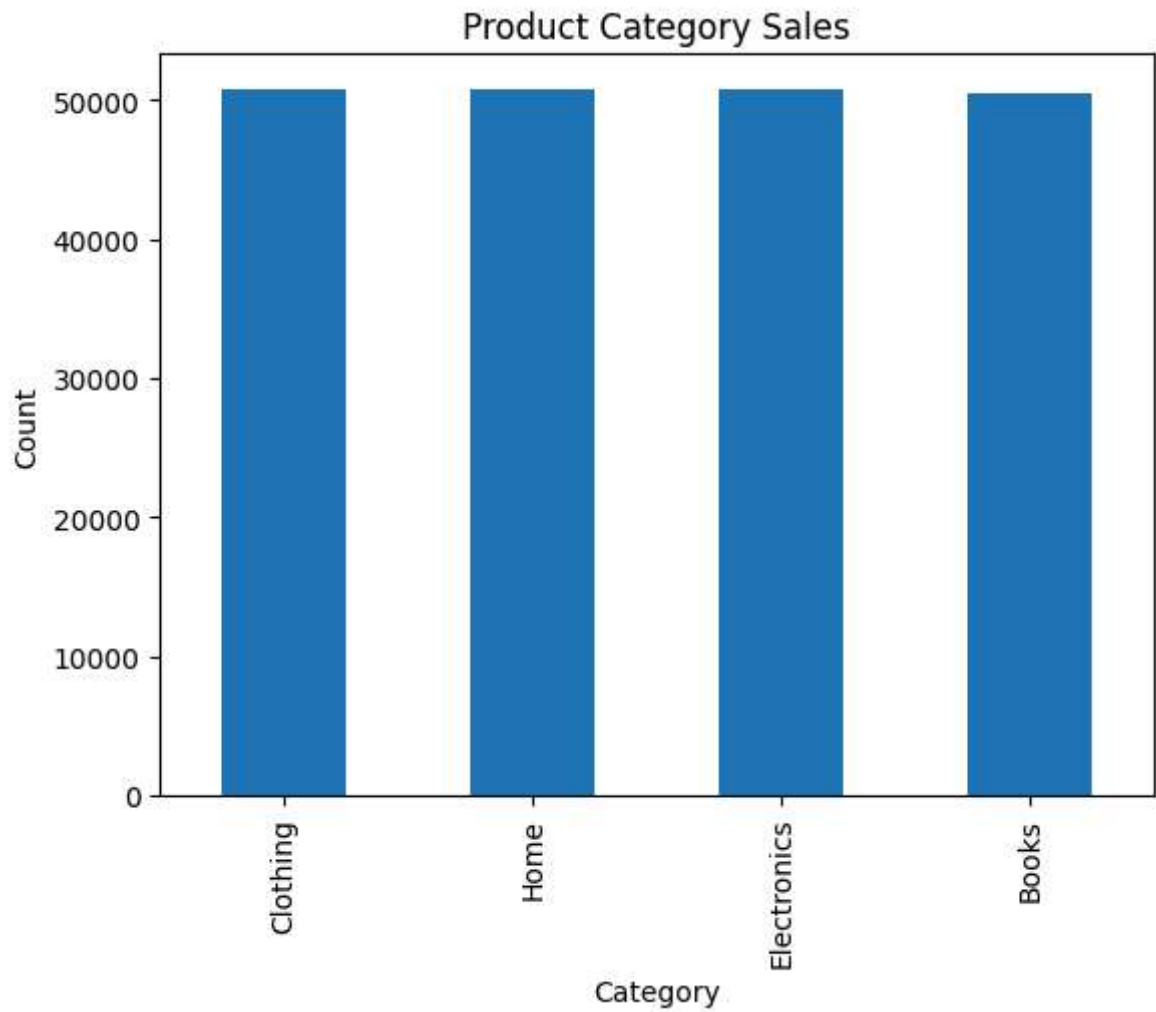
plt.title("Customer Segments")
plt.show()
```



```
In [28]: df['Product Category'].value_counts().plot(kind='bar')

plt.title("Product Category Sales")
plt.xlabel("Category")
plt.ylabel("Count")
```

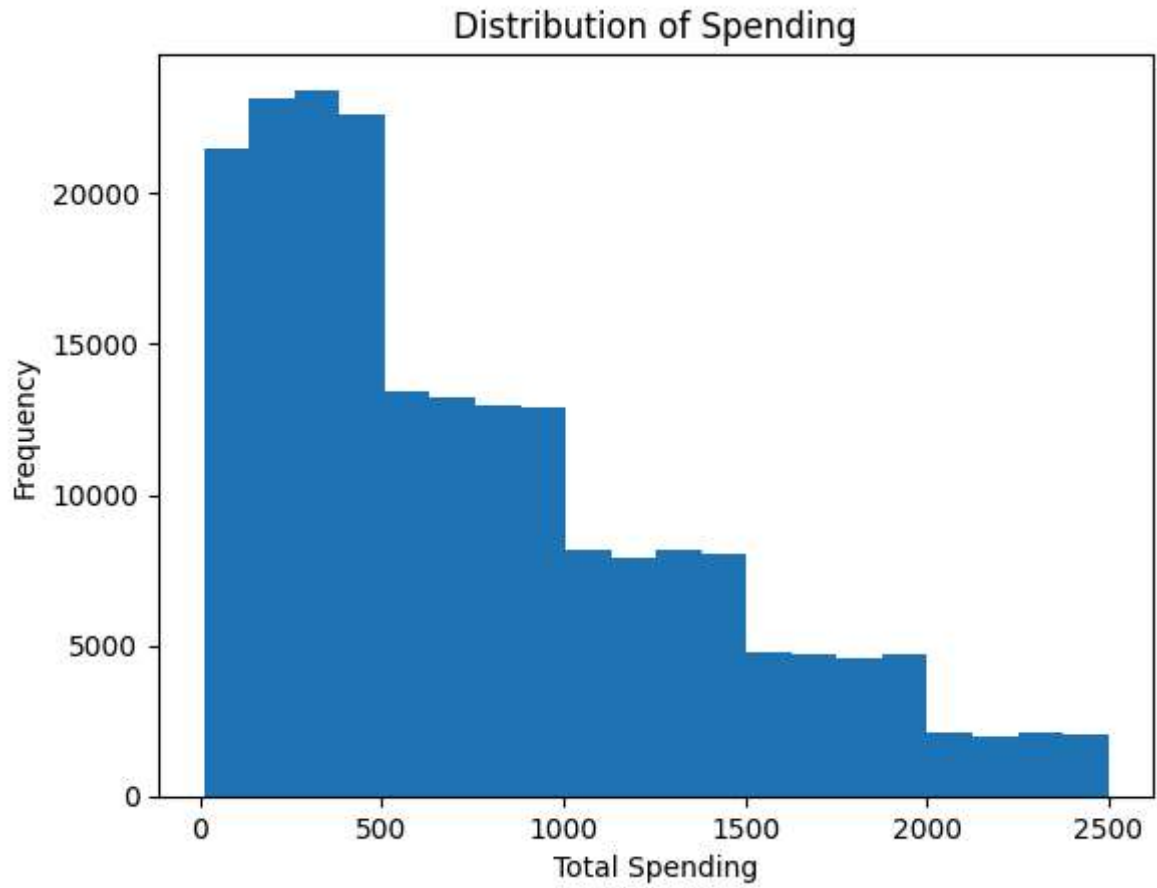
```
plt.show()
```



```
In [29]: plt.hist(df['Total_Spending'], bins=20)

plt.title("Distribution of Spending")
plt.xlabel("Total Spending")
plt.ylabel("Frequency")

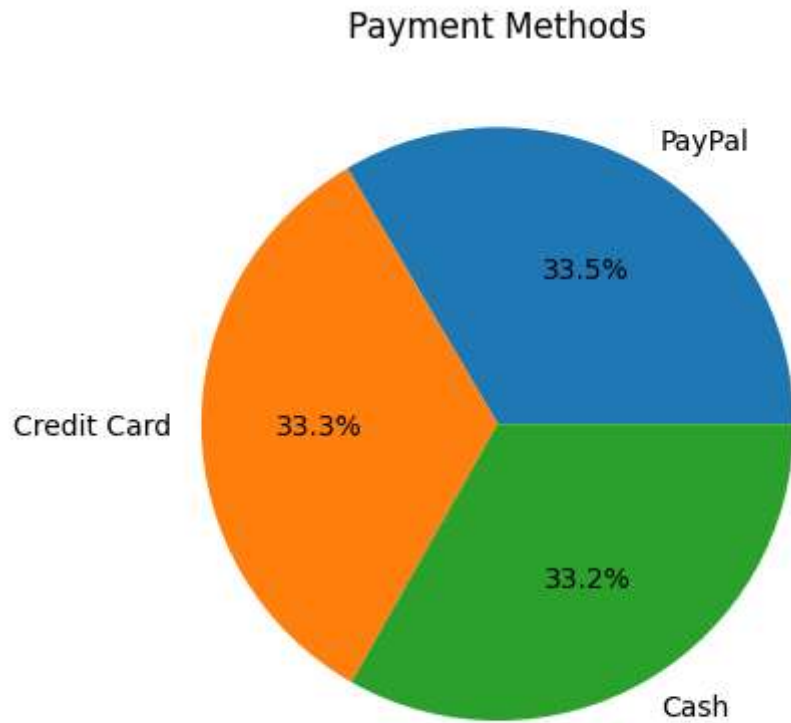
plt.show()
```



```
In [30]: df['Payment Method'].value_counts().plot(
        kind='pie',
        autopct='%1.1f%%'
    )

plt.title("Payment Methods")

plt.show()
```



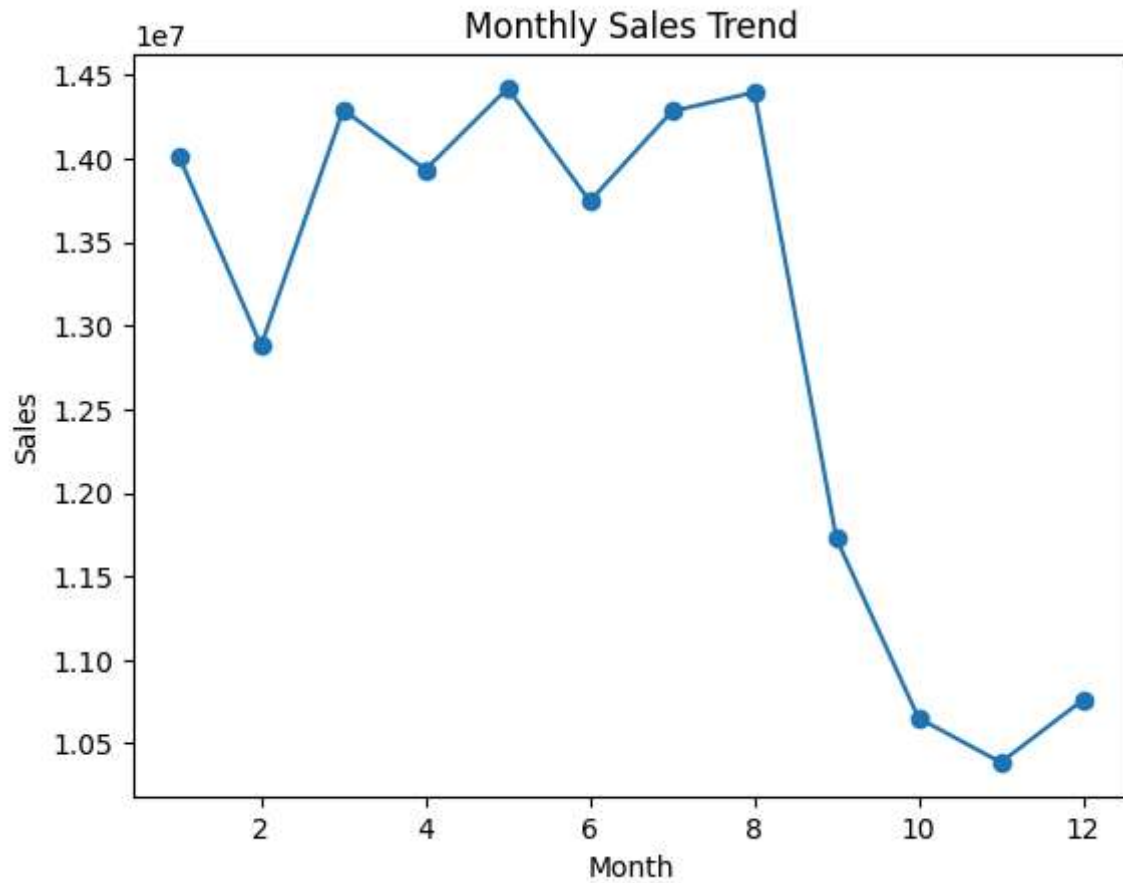
```
In [31]: df['Month'] = df['Purchase Date'].dt.month

monthly_sales = df.groupby('Month')['Total_Spending'].sum()

monthly_sales.plot(kind='line', marker='o')

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.show()
```

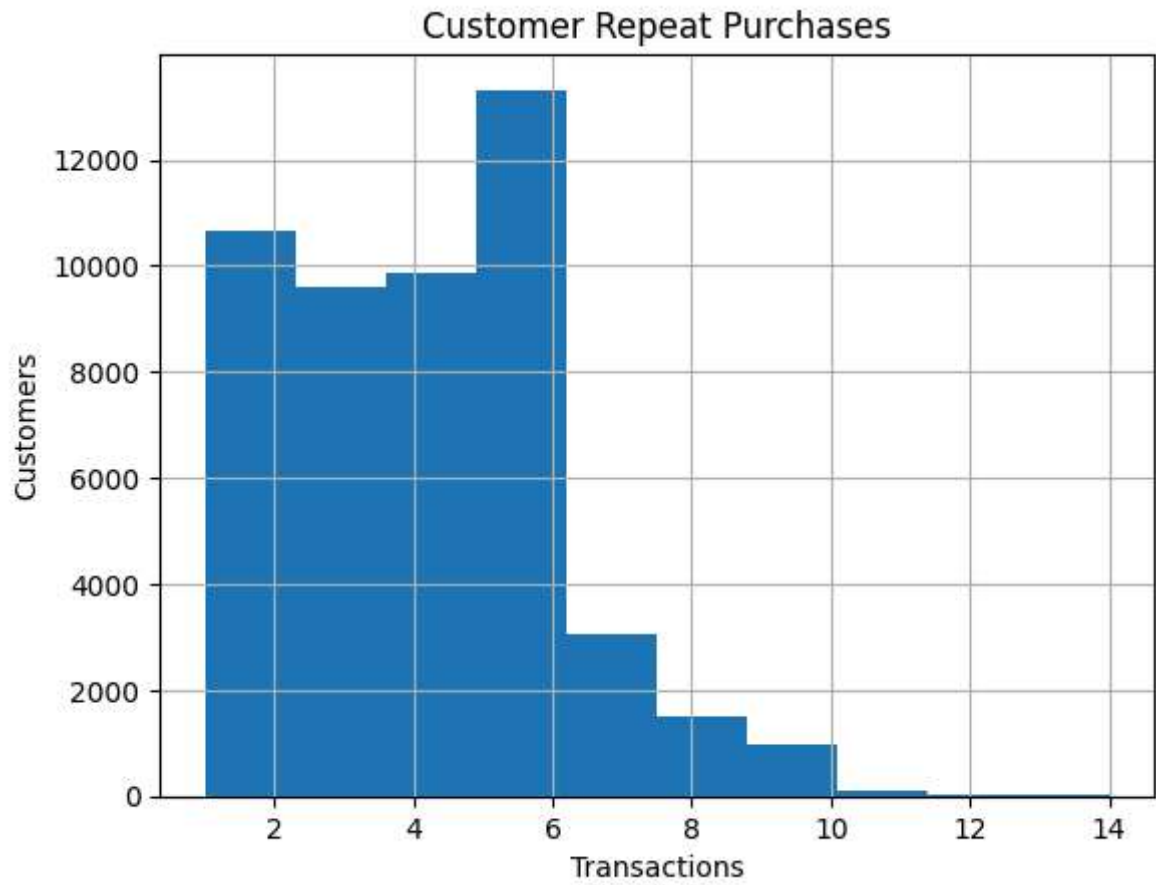


```
In [32]: repeat_customers = df.groupby('Customer ID').size()

repeat_customers.hist()

plt.title("Customer Repeat Purchases")
plt.xlabel("Transactions")
plt.ylabel("Customers")

plt.show()
```



In []: